

Hemish Jain

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Experience

Machine Learning Intern | JA Assure | Remote January 2026 – March 2026

- Built a three-layer end-to-end pipeline to convert unstructured PDF documents into structured data: Layer 1 used vision models with prompt tuning to extract text, tables, and checkbox data; Layer 2 used an LLM to structure output into standardized JSON; Layer 3 automated spreadsheet population via API.
- Designed the entire system to run fully offline using open-source models on company hardware, ensuring sensitive data privacy while significantly reducing manual effort in high-volume document workflows.

Publications

Suicidal Text Detection Using Machine Learning and Large Language Models [PDF](#) | [GitHub](#)

- Developed a suicidal text detection system using Reddit data (r/SuicideWatch, r/depression, r/teenagers), comparing traditional ML models with a fine-tuned BERT to identify suicidal ideation with high accuracy.
- BERT significantly outperformed Gaussian Naive Bayes and Logistic Regression, achieving 97.68% accuracy; published at ICRAIS 2025, MIT Manipal.

Multimodal Anemia Detection Using Convolution Neural Network and Ensemble Learning(Accepted at CVIP 2025)

- Developed a multimodal anemia-detection system that fuses palm-image CNNs (MobileNetV2, ResNet50, EfficientNetB0) with CBC-based ML models (Logistic Regression, Random Forest, SVM) in a two-level stacked ensemble using soft voting and a meta-learner, achieving 93.57% accuracy and 95.50% recall.
- Designed for low-resource, real-time deployment in rural and under-resourced healthcare settings, enabling non-invasive early screening using both visual and clinical indicators.

Projects

Suicidal Text Detection Bot (Python, Transformers, Scikit-learn, Hugging Face) [Live Project](#)

- Developed an NLP-based model to detect suicidal ideation using BERT, Gaussian Naive Bayes, and Logistic Regression.
- Integrated Hugging Face Hub APIs for real-time inference and deployed on Vercel using Flask backend.

Reel & Blog Summarizer (Python, Gemini API, Speech-to-Text, SQLite) [Github Repo](#)

- Built a personal productivity tool that processes YouTube/reel videos and blogs: extracts transcripts (or downloads and transcribes via speech-to-text if unavailable), then sends content to Gemini API to generate concise summaries with keywords.
- Summaries are stored in a structured database and indexed for fast keyword-based and contextual retrieval, significantly reducing time spent re-locating previously saved content.

Technologies

Languages: Python, Java, C++, C, SQL, JavaScript

Frameworks & Tools: PyTorch, Scikit-learn, Flask, Spring Boot, MySQL, Vercel, Hugging Face, TensorFlow/Keras, OpenCV, Pandas, NumPy, Matplotlib, Bootstrap

Areas of Interest: Machine Learning, NLP, Computer Vision, IoT

Education

B.Tech in Computer Science and Engineering | SRM IST, Chennai 2024 – 2028

GPA: 8.3/10

Certifications

Oracle OCI Data Science Foundations • AWS Cloud Foundations • AWS Cloud Architecting • Tata Strive Cybersecurity